EXP 21:

Perform Sharpening of Image using Laplacian mask implemented with an extension of diagonal neighbors,

1	**1**	**1**
1	**-8**	**1**
1	**1**	**1**

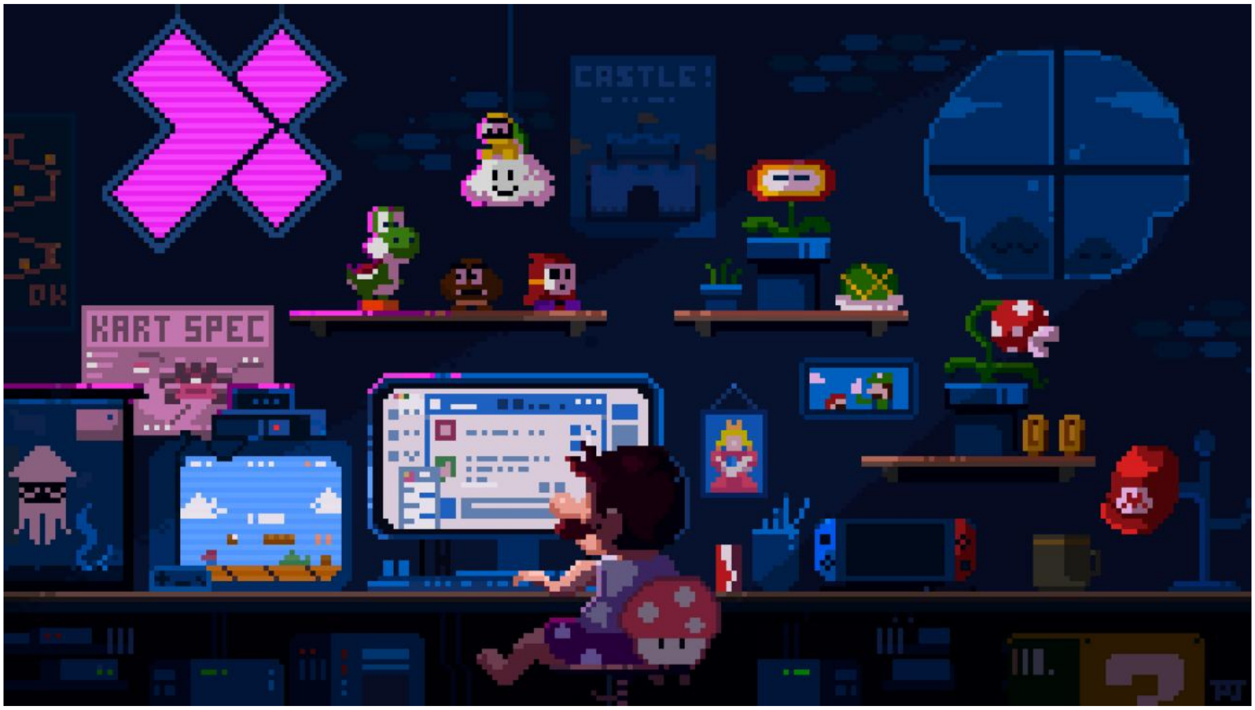
AIM:

Perform Sharpening of Image using Laplacian mask implemented with an extension of diagonal neighbors..

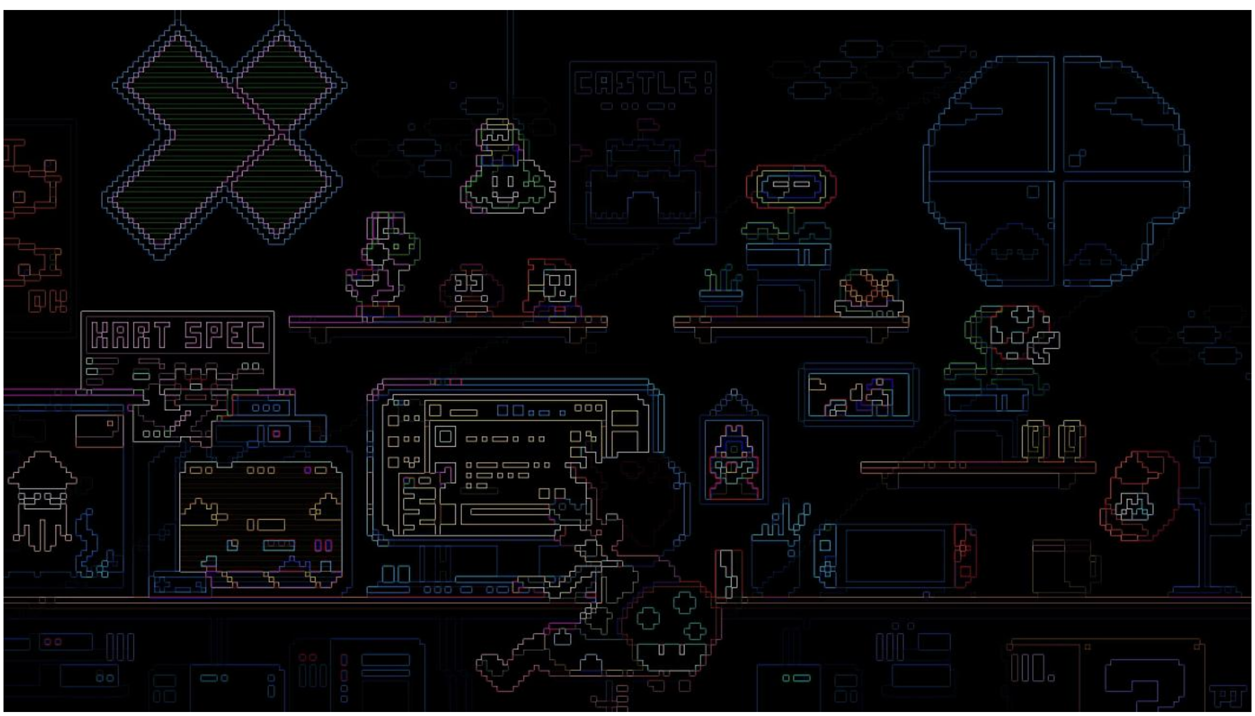
Program:

```
import cv2

import numpy as np

img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

else:

    kernel = np.array([[1, 1, 1],

                       [1, -8, 1],

                       [1, 1, 1]])

    sharpened = cv2.filter2D(img, -1, kernel)


    cv2.imshow('Original Image', img)

    cv2.imshow('Sharpened Image (Diagonal Laplacian)', sharpened)


    cv2.imwrite('Sharpened_Image_Exp21.jpg', sharpened)


    cv2.waitKey(0)

    cv2.destroyAllWindows()
```

INPUT:



Output:



RESULT:- Thus, the program is executed successfully.